执行数据库备份

Snap Creator Framework

NetApp
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执行数据库备份

您可以使用 Snap Creator GUI 或命令行备份 SAP HANA 数据库。要计划备份，您可以在 GUI 中使用计划程序，也可以将命令行与 cron 等外部计划程序结合使用。

数据库备份概述

在 Snap Creator 备份数据库时，将执行以下步骤。

1. 创建一个全局同步备份保存点（ SAP HANA Snapshot 副本），以便在持久性层上获得一致的映像。
2. 为所有数据卷创建存储 Snapshot 副本。

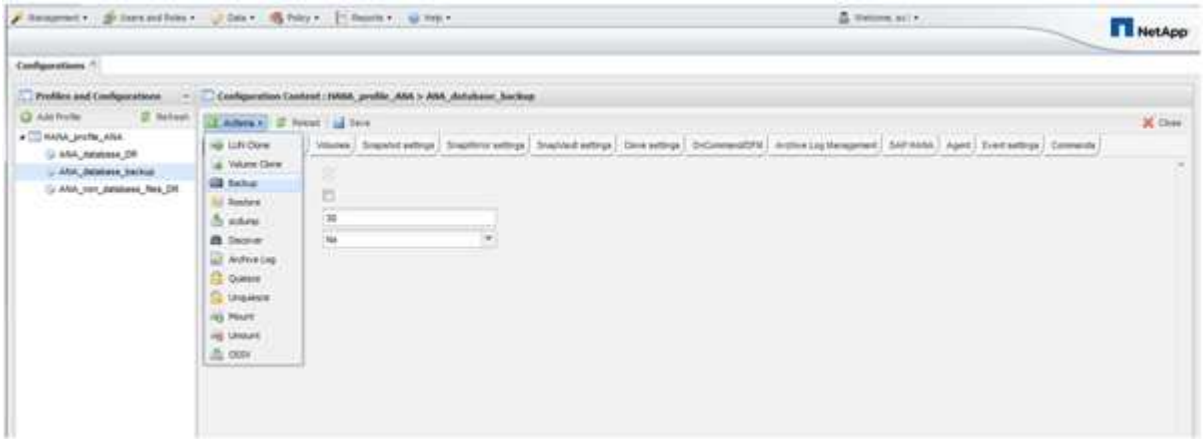
   在此示例中，有三个数据卷分布到两个存储控制器： hana1a 和 hana1b 。

3. 在 SAP HANA 备份目录中注册存储 Snapshot 备份。
4. 删除 SAP HANA Snapshot 副本。
5. 为所有数据卷启动 SnapVault 更新。
6. 检查 SnapVault 状态并等待，直到超时完成或可配置为止。
7. 根据为主存储和二级存储上的备份定义的保留策略，删除 SAP HANA 备份目录中的存储 Snapshot 副本和备份。
8. 删除文件系统和 SAP HANA 备份目录中早于最早数据备份的所有日志备份。

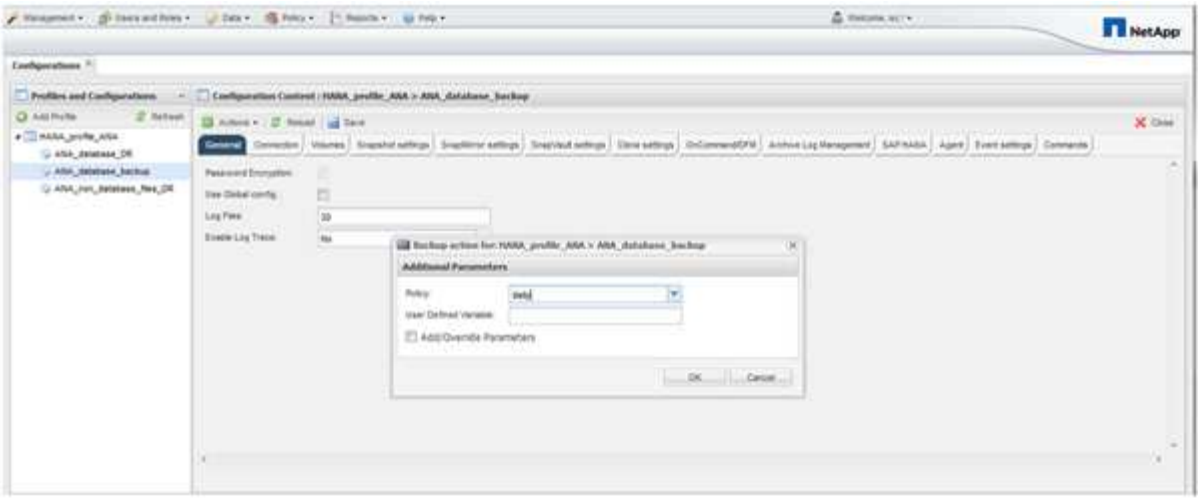
使用 **Snap Creator GUI** 备份数据库

您可以使用 Snap Creator GUI 备份数据库。

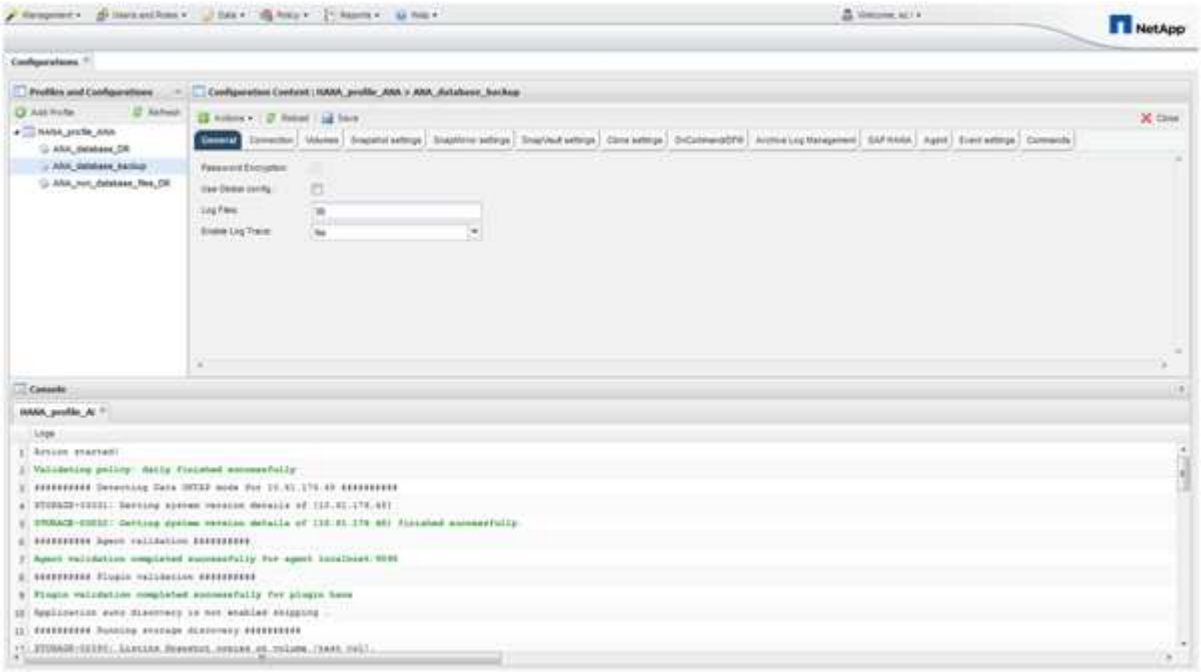
1. 选择 * HANA 数据库备份配置 * ，然后选择 * 操作 * > * 备份 * 。



2. 选择备份策略并单击 * 确定 * 。



备份将开始。Snap Creator 会触发 SnapVault update ，` ， Snap Creator 会等待数据复制到二级存储。此等待时间已在配置期间进行配置，可在 SnapVault 设置选项卡中进行调整。对于同一存储控制器上的每个卷， Snap Creator 会同时触发 SnapVault 更新，但对于每个存储控制器，此更新会按顺序触发。



使用 **Snap Creator** 命令行备份数据库

您也可以使用 Snap Creator 命令行备份数据库。

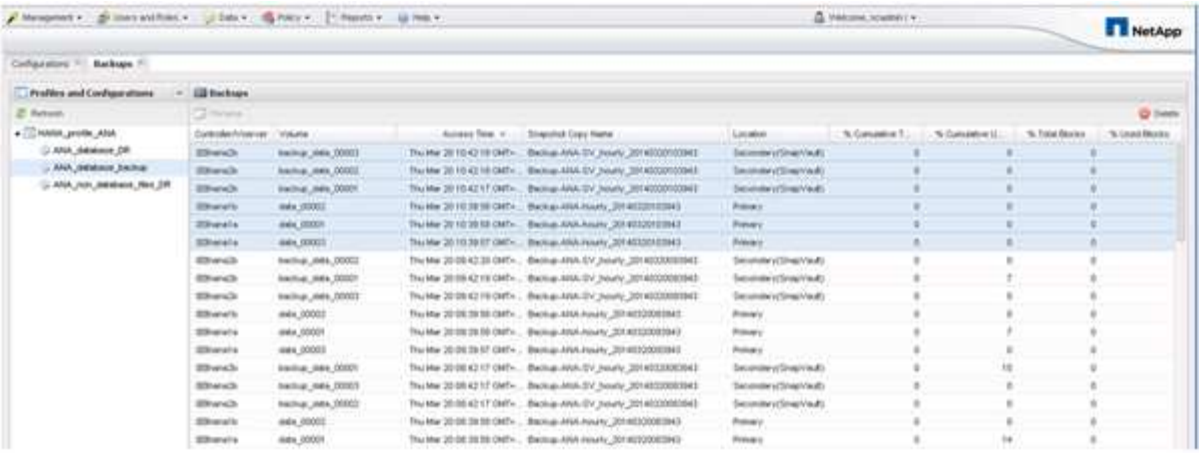
1. 要备份数据库，请运行以下命令。

```
mgmtsrv01:~ #
/opt/NetApp/Snap_Creator_Framework_411/scServer4.1.1/snapcreator
--server
localhost --port 8443 --user scadmin --passwd scadmin --profile
HANA_profile_ANA --config
ANA_database_backup --action backup --policy daily --verbose
[Wed Mar 5 14:17:08 2014] INFO: Validating policy: daily finished
successfully

########## Detecting Data ONTAP mode for hana1a ##########

########## Detecting Data ONTAP mode for hana1b ##########
[Wed Mar  5 14:17:13 2014] INFO: STORAGE-03031: Getting system version
details of [hana2b]
[Wed Mar  5 14:17:13 2014] INFO: STORAGE-03032: Getting system version
details of [hana2b] finished successfully.
[Wed Mar  5 14:17:13 2014] INFO: STORAGE-03031: Getting system version
details of [hana1a]
[Wed Mar  5 14:17:13 2014] INFO: STORAGE-03032: Getting system version
details of [hana1a] finished successfully.
[Wed Mar  5 14:17:13 2014] INFO: STORAGE-03031: Getting system version
details of [hana1b]
[Wed Mar  5 14:17:13 2014] INFO: STORAGE-03032: Getting system version
details of [hana1b] finished successfully.

….
Truncated
…
```

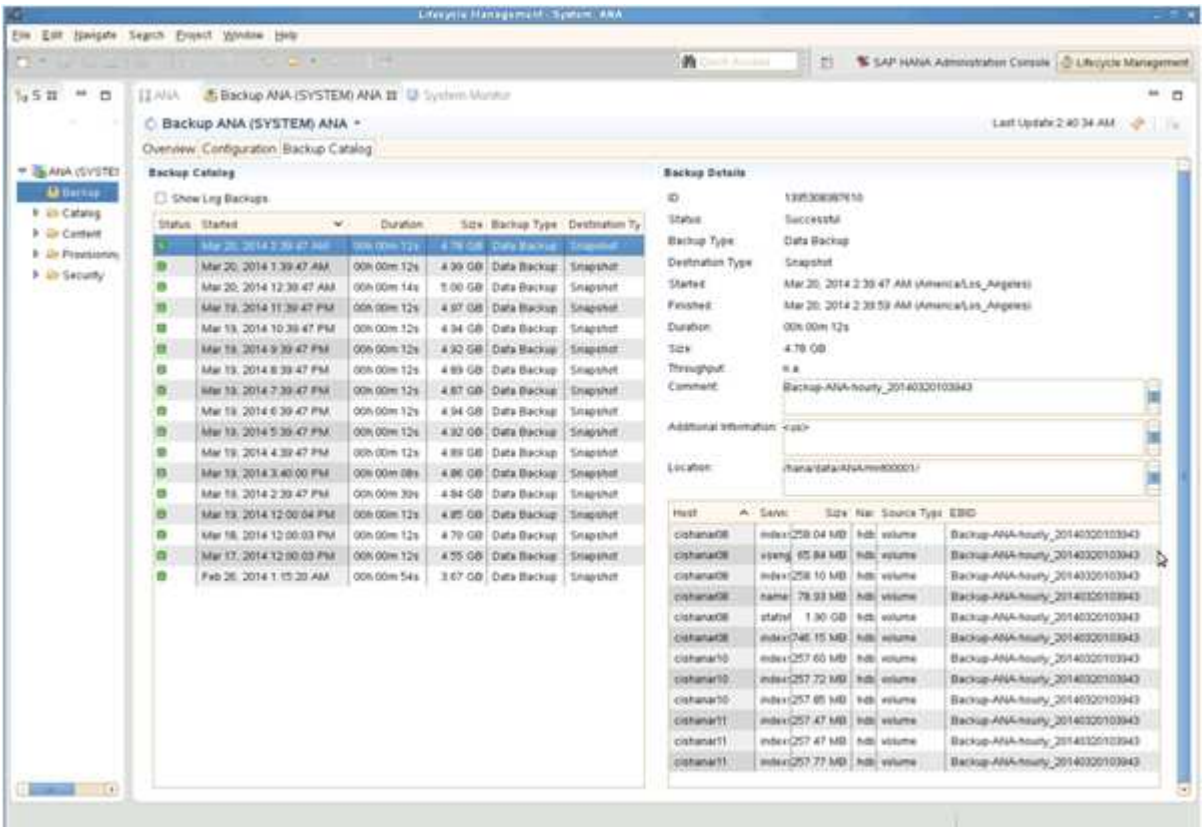
在 SAP HANA Studio 中查看可用备份

您可以在 SAP HANA Studio 中查看存储 Snapshot 备份列表。

下图中突出显示的备份显示了一个名为 "`Backup-ANA _hourly_20140320103943 的 Snapshot 副本。` 此备份包括 SAP HANA 系统所有三个数据卷的 Snapshot 副本。备份也可从二级存储获得。



当 Snap Creator 在 SAP HANA 备份目录中注册存储 Snapshot 副本时， Snap Creator 会使用 Snapshot 副本名称作为备份 ID 。在 SAP HANA Studio 中，存储 Snapshot 备份会显示在备份目录中。外部备份 ID （ EBID ）与 Snapshot 副本名称的值相同，如下图所示。



每次运行备份时， Snap Creator 都会根据为不同计划（每小时，每天等）定义的保留策略删除主存储和二级存储上的 Snapshot 备份。

如果主存储或二级存储不存在备份，则 Snap Creator 还会删除 SAP HANA 备份目录中的备份。SAP HANA 备份目录始终显示主存储和 / 或二级存储上可用的备份的完整列表。

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